

SEHAJDEEP SINGH

+1 (647) 987 - 0490 | sehajdeep490@yahoo.com | linkedin.com/in/singh-sehaj-deep | github.com/sehaj-deep

EDUCATION

Concordia University || Montreal, Canada Jan 2024 – Aug 2025

Master of Applied Computer Science (GPA - 4.17 /4.3)

Relevant Coursework: Data Structures and Algorithms, Programming Practices, Distributed Systems

DAV Institute of Engineering and Technology || Jalandhar, India

Jul 2019 – Jun 2023

Bachelor of Technology, Computer Science (GPA - 3.59/ 4.0)

Relevant Coursework: Operating Systems, Database Management, Compiler Design

SKILLS

- **Areas of Interest:** AI & ML, NLP, Continual Learning, Active Learning, Backend Development, Algorithms
- **Programming Languages:** Python (Proficient), Java(Proficient), C++, C, JavaScript, SQL
- **Backend:** Spring Boot 3.2, Spring Security, REST APIs, JUnit, Maven, Microservices
- **Cloud/DevOps:** AWS (EC2, S3, DynamoDB, SQS, Lambda), Docker, CI/CD, GitHub Actions, Jenkins, Git
- **Environments & IDEs:** Linux/UNIX, Windows, IntelliJ, VS Code, Eclipse

WORK EXPERIENCE

Computer Science Teaching Assistant || Concordia University, Montreal, Canada Sep 2024 - Aug 2025

- Mentored 200+ students across Data Structures, Programming Languages, and AI courses at Concordia University, leading tutorials and labs while collaborating with faculty to improve learning outcomes.

Software Developer || We.PLM Pvt. Limited, Pune, India

Jul 2023 - Dec 2023

- Led design and development of automated carbon neutrality dashboard, integrating product data from 5+ teams to contribute to 45% carbon emissions reductions toward 2027 target.
- Built scalable Java backend with RESTful APIs and Jenkins pipelines for real-time analytics dashboard, delivering sprints 15% ahead of schedule through optimized data syncing.

Software Engineering Intern || STMicroelectronics Pvt. Limited, Greater Noida, India

Jul 2022 – Jun 2023

- Developed automated testing frameworks using Selenium WebDriver, Selenide, Cucumber, and TestNG, reducing regression testing time from 3 days to 1 day for critical software releases.
- Coordinated with international teams including Dassault Systèmes and Capgemini partners, managing multiple priorities across time zones to deliver software features on schedule for production releases.

PROJECT EXPERIENCE

AuthBridge - Enterprise Identity Provider || [Github](https://github.com)

- Developed Java 17/Spring Boot 3.2 service with Spring Security, REST APIs, LDAP bind auth (UnboundID SDK), and JWT SSO enabling enterprise single sign-on.
- Containerized OpenLDAP with Docker simulating Active Directory for delegated authentication across services.
- Implemented stateless JWT validation eliminating per-request DB lookups supporting scalable distributed authentication architecture.

Movie Recommendation System

- Built and documented distributed PySpark engine processing 5.5M+ reviews on AWS (EC2, S3, DynamoDB, SQS, Lambda) and Docker delivering top-10 movie recommendations.
- Implemented EC2 scaling with master-worker election and SQS decoupling enabling concurrent processing across multiple instances.
- Deployed DynamoDB Global Tables across 2 regions ensuring regional failover and fault tolerance.

Command - Line Risk Game || [Github](https://github.com)

- Architected Java Risk game using MVC architecture and design patterns (Observer, Strategy, Command, Adapter) with 95% JUnit test coverage through TDD for robust multi-player tournament testing.
- Implemented GitHub Actions CI/CD pipeline with SonarQube vulnerability scanning, Maven builds, and led code reviews across 5-developer sprints to ensure production-ready code quality.